YEAR 11 PHYSICS, AQA GCSE, LESSON 3 OF 10

Atoms & Nuclear Radiation.

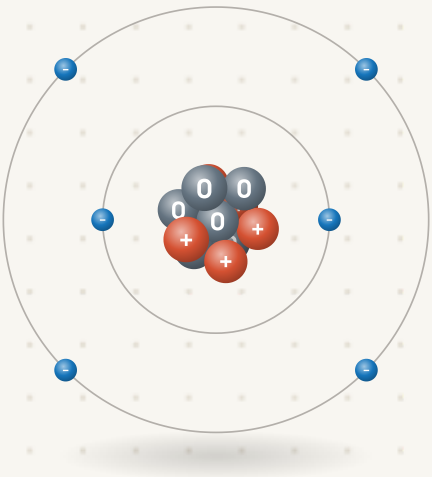
Lesson 03 From plum pudding to the nucleus

Nobody has ever seen inside an atom. So how did scientists find out that almost all of its mass is squeezed into a tiny nucleus?

IN THIS LESSON YOU WILL

- describe the plum pudding model and why it was proposed
- describe the alpha particle scattering experiment
- explain how each result led to the nuclear model
- compare the plum pudding and nuclear models

ANSWER EDITION FOR TEACHERS AND SELF-MARKING



NAME _____ CLASS _____

TEACHER _____ TARGET GRADE _____

START HERE

How this booklet works.

This is the booklet for **Lesson 3** of the Atoms and Nuclear Radiation unit. It builds on the lessons before it; the "Do now" questions and the quick answers at the back help you check what you remember. Keep it: it is your classwork book *and* your revision guide.

DO NOW
Quick questions from earlier lessons. Remembering again is what makes it stick.

LEARN IT
Explanations, diagrams and worked examples. Read this again when you revise.

TRY IT
Practice that builds up: first with support, then on your own, then a challenge.

LAB
Record what you find in a simulation or a model experiment.

EXAM CORNER
AQA-style questions with a tip on how the marks are given.

REVISE IT
Summary, cover-and-answer questions and an "I can" check to colour in.

! **Revising at home?** (1) Read the Learn it pages. (2) Cover them and answer the "Cover the page" questions in Revise it. (3) Check with the quick answers at the back. (4) Go through the matching slides in the teaching deck, and try the simulations.

The route through the unit (this lesson is highlighted)

01 Inside the atom

02 Isotopes and ions

03 From plum pudding to the nucleus
The plum pudding model, alpha scattering and the nuclear model (slides 9–17)

04 Energy levels and the nucleus

05 Radioactive decay

06 Nuclear equations

07 Properties and uses of radiation

08 Half-life and random decay

09 Half-life calculations

10 Contamination and irradiation

Teaching deck: **panphy.app/gcsephy/decks/atoms-and-radiation**. Simulations: **panphy.app/simulations/atomic_models.html** and **panphy.app/simulations/nuclear_decay.html**

LESSON

03 From plum pudding to the nucleus.

AQA Physics 4.4.1.3; Combined Science 6.4.1.3

BIG QUESTION

Nobody has ever seen inside an atom. So how did scientists find out that almost all of its mass is squeezed into a tiny nucleus?

BY THE END YOU CAN

- describe the plum pudding model and why it was proposed
- describe the alpha particle scattering experiment
- explain how each result led to the nuclear model
- compare the plum pudding and nuclear models

model | plum pudding model | alpha particle | alpha scattering | deflected | nuclear model | evidence

SLIDES AND SIMULATIONS

Teaching deck, slides 9–17

Timeline of models, the plum pudding model, alpha scattering and the nuclear model.

panphy.app/gcsephy/decks/atoms-and-radiation/#9

Atomic Models simulation

panphy.app/simulations/atomic_models.html

RETRIEVE

Do now

From earlier lessons. No notes!

1 What particles are in the nucleus?

Protons and neutrons.

2 What is the overall charge of a nucleus? Why?

Positive: protons are positive and neutrons have no charge.

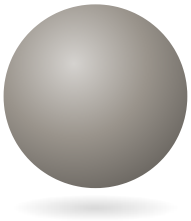
3 The radius of a nucleus is less than what fraction of the atom's radius?

1/10 000

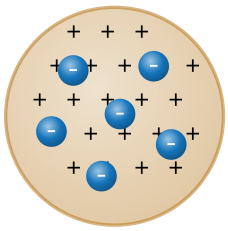
EXPLAIN

How the model changed

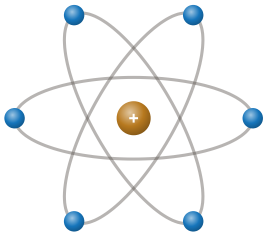
A model that kept changing



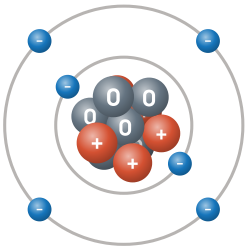
Before 1897
Tiny solid spheres that cannot be divided



1904
Plum pudding: after the electron was discovered



1911
Nuclear model: from alpha scattering



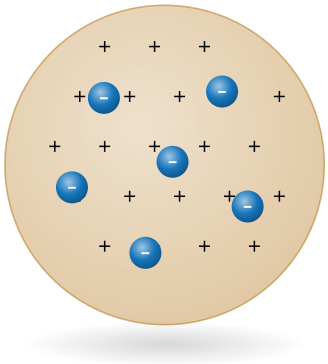
1913
Bohr model: electrons in energy levels

Scientists use **models** to explain things too small to see. When new **evidence** does not fit a model, the model is changed or replaced.

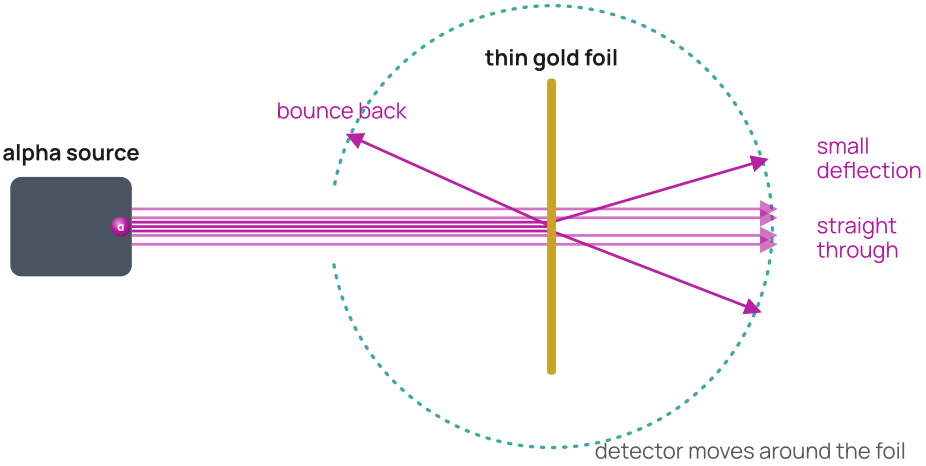
The plum pudding model (1904)

In 1897 J. J. Thomson discovered the **electron**, a tiny negative particle that comes from inside atoms. So atoms *can* be divided into smaller parts.

This led to the **plum pudding model**: the atom is a **ball of positive charge** with **negative electrons embedded in it**. The positive charge and the mass are spread through the whole atom. There is **no nucleus**.



The alpha scattering experiment (1909)



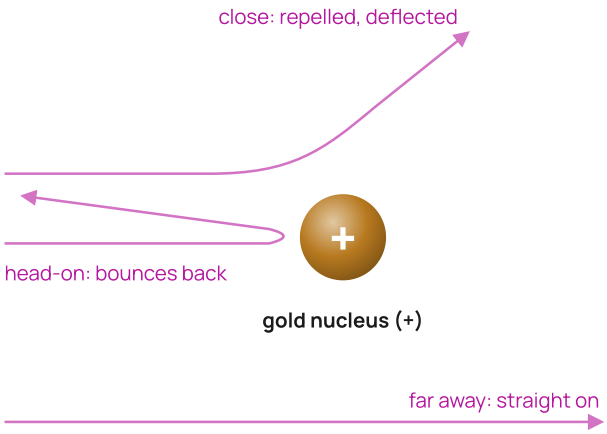
Geiger and Marsden, working with Rutherford, fired alpha particles at a very thin gold foil and counted them at every angle.

Alpha particles are small, fast and positively charged. The plum pudding model predicted that they would all pass **straight through** the foil, with at most tiny deflections, because the

positive charge was spread out too thinly to push them far off course.

What they saw, and what it means

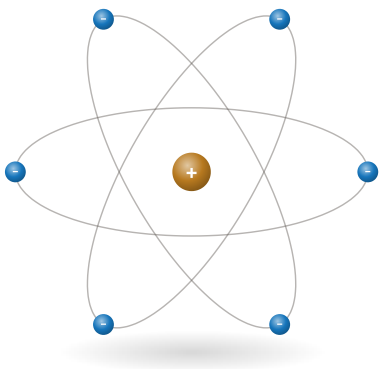
Observation	Conclusion
Most alpha particles went straight through.	The atom is mostly **empty space**.
Some were deflected through large angles.	The centre of the atom is **positively charged**: it repels the positive alpha particles.
A very few bounced back.	The **mass** and charge are concentrated in a **tiny centre**: the nucleus.



Close-up of one gold nucleus: not to scale.

The nuclear model (Rutherford, 1911)

	Plum pudding	Nuclear model
Positive charge	spread through the atom	in a tiny nucleus
Mass	spread through the atom	concentrated in the nucleus
Empty space?	no	mostly empty space
Electrons	embedded in the positive ball	outside the nucleus



! The plum pudding model could not explain alpha particles bouncing back. The new evidence did not fit, so the model was **replaced** by the nuclear model.

PRACTISE

Try it

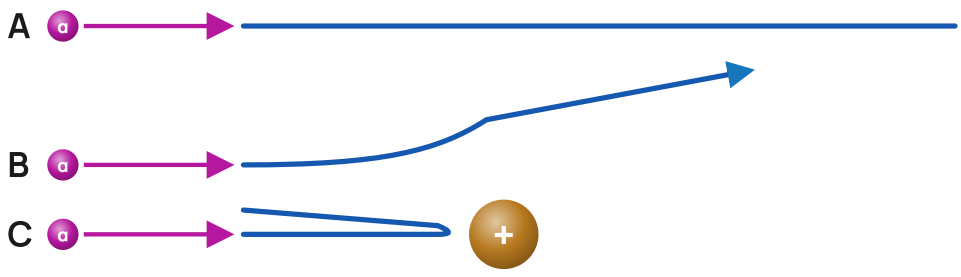
1 Put the models of the atom in the order they were proposed. Write 1 (first) to 4 (latest) in the boxes.

3	A tiny, positive nucleus with electrons outside it
1	Tiny solid spheres that cannot be divided
4	Electrons orbit the nucleus at fixed distances (energy levels)
2	A ball of positive charge with electrons embedded in it

2 Draw a line from each observation to the conclusion it led to. (The answer edition shows the matching letter.)

1 Most alpha particles passed straight through.	→ B	A The mass is concentrated in a tiny centre.
2 Some alpha particles were deflected.	→ C	B Most of the atom is empty space.
3 A very few alpha particles bounced back.	→ A	C The centre of the atom has a positive charge.

3 Three alpha particles, A, B and C, move towards a gold nucleus. Continue the path of each one.



gold nucleus

A passes far from the nucleus, B passes close to it and C is heading straight for it.

A carries straight on. B is repelled and deflected away from the nucleus. C slows, stops and bounces back the way it came.

4 Explain why the plum pudding model cannot explain alpha particles bouncing back.

In the plum pudding model the positive charge and mass are spread out through the atom. That is not concentrated enough to repel a fast, positive alpha particle strongly enough to turn it back. Bouncing back needs a tiny, dense, positive centre.

5 Is Mia right? Explain your answer.

> **Mia:** Most alpha particles went straight through, so the experiment proved the plum pudding model was right.

No. The plum pudding model also predicts that most go straight through, so that result does not decide between the models. The large deflections and the bounce-backs did not fit the plum pudding model, so it was replaced.

6 CHALLENGE The gold foil was only a few hundred atoms thick. Suggest why a very thin foil was needed.

Alpha particles are easily stopped, so they need thin foil to pass through at all. Each one then meets only a few nuclei, so each large deflection comes from one close approach to a single nucleus.

AQA-STYLE

Exam corner

Write in full sentences. Show every step of a calculation.

Get the marks: In a 'compare' question, say what each model says about the same feature: charge, mass, empty space, electrons.

7 Describe the differences between the plum pudding model and the nuclear model of the atom. [4 marks]

Any four: plum pudding has the positive charge spread out; nuclear model has it in a tiny nucleus (1). Mass spread out vs concentrated in the nucleus (1). Plum pudding has no empty space; the nuclear atom is mostly empty space (1). Electrons embedded in the positive charge vs electrons outside the nucleus (1).

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8 In the alpha scattering experiment, most alpha particles passed straight through the gold foil, but a very small number were deflected by more than 90°. Explain what each result showed about the structure of the atom. **[4 marks]**

Most passing straight through shows the atom is mostly empty space (2). A very small number deflected by more than 90° shows there is a tiny centre (1) where the positive charge and mass are concentrated, which repels alpha particles (1).

REVIEW LATER

Revise it

Use this box after the lesson and before tests.

SUMMARY

- Models change when new evidence does not fit them.
- Plum pudding (1904): a ball of positive charge with electrons embedded; no nucleus.
- Alpha scattering: most straight through (empty space), some deflected (positive centre), a very few bounced back (mass in a tiny centre).
- Nuclear model (1911): tiny, dense, positive nucleus with electrons outside.
- The nuclear model replaced the plum pudding model.

COVER THE PAGE AND ANSWER

1 What discovery led to the plum pudding model?

2 What were alpha particles fired at?

3 What did bounce-backs show?

4 Where is the positive charge in the nuclear model?

I CAN…

- describe the plum pudding model
- describe the alpha scattering experiment
- link each result to a conclusion about the atom
- compare the plum pudding and nuclear models

REFERENCE

Toolkit.

The facts and methods you have met so far. Learn these: exam questions expect you to recall them without a data sheet.

The three particles

Lesson 1

Particle	Charge	Mass	Where
Proton	+1	1	nucleus
Neutron	0	1	nucleus
Electron	−1	very small	energy levels

Reading the symbol

Lesson 1

$^{A}_{Z}X$ e.g. $^{23}_{11}Na$

- Z, atomic number = protons
- A, mass number = protons + neutrons
- neutrons = A − Z; in an atom, electrons = Z
- Atom radius ≈ 1×10^{-10} m; nucleus radius < 1/10 000 of that

Isotopes and ions

Lesson 2

- **Isotopes**: same number of protons, different numbers of neutrons.
- **Positive ion**: an atom that has lost one or more outer electrons. The nucleus is unchanged.

Alpha scattering

Lesson 3

Observation	Conclusion
most straight through	mostly empty space
some deflected	positive centre
a very few bounced back	mass in a tiny centre

REFERENCE

Key words from this lesson.

Learn the meaning and spelling. In exams, precise words earn marks: say “nucleus”, not “middle”, and “unstable nucleus”, not “unstable atom”.

Alpha scattering: The experiment in which alpha particles fired at thin gold foil showed that atoms have a tiny, positive nucleus.

Evidence: Observations and measurements used to test a model.

Model: A simplified explanation of something we cannot see directly; changed when new evidence does not fit.

Nuclear model: A tiny, dense, positive nucleus with electrons outside it (Rutherford, 1911).

Plum pudding model: A ball of positive charge with negative electrons embedded in it (1904).

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CHECK YOURSELF

Quick answers.

Answers to the “Do now” and “Cover the page” questions, so you can check yourself when revising. Answers to the longer questions are in the answer edition.

DO NOW

1. Protons and neutrons.
2. Positive: protons are positive and neutrons have no charge.
3. 1/10 000

COVER THE PAGE

1. The electron (atoms can be divided).
2. Very thin gold foil.
3. The mass (and positive charge) is concentrated in a tiny centre, the nucleus.
4. In a tiny nucleus at the centre.

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